

WEEK 7

Date: 10/09/2026

Title: Method Overriding using RBI and Different Banks

Objective / Aim

To design and implement a Java program to demonstrate **method overriding and runtime polymorphism** using RBI as the parent class and SBI, ICICI, and PNB as child classes with different rates of interest.

System Requirements

Hardware

- Standard PC/Laptop
- Minimum 4 GB RAM
- Minimum 500 MB free storage

Software

- Operating System: Windows / Linux / macOS
 - Java Development Kit (JDK 17 or later)
 - Text Editor or IDE such as VS Code, Eclipse, IntelliJ IDEA, or NetBeans
 - Command Prompt / Terminal
-

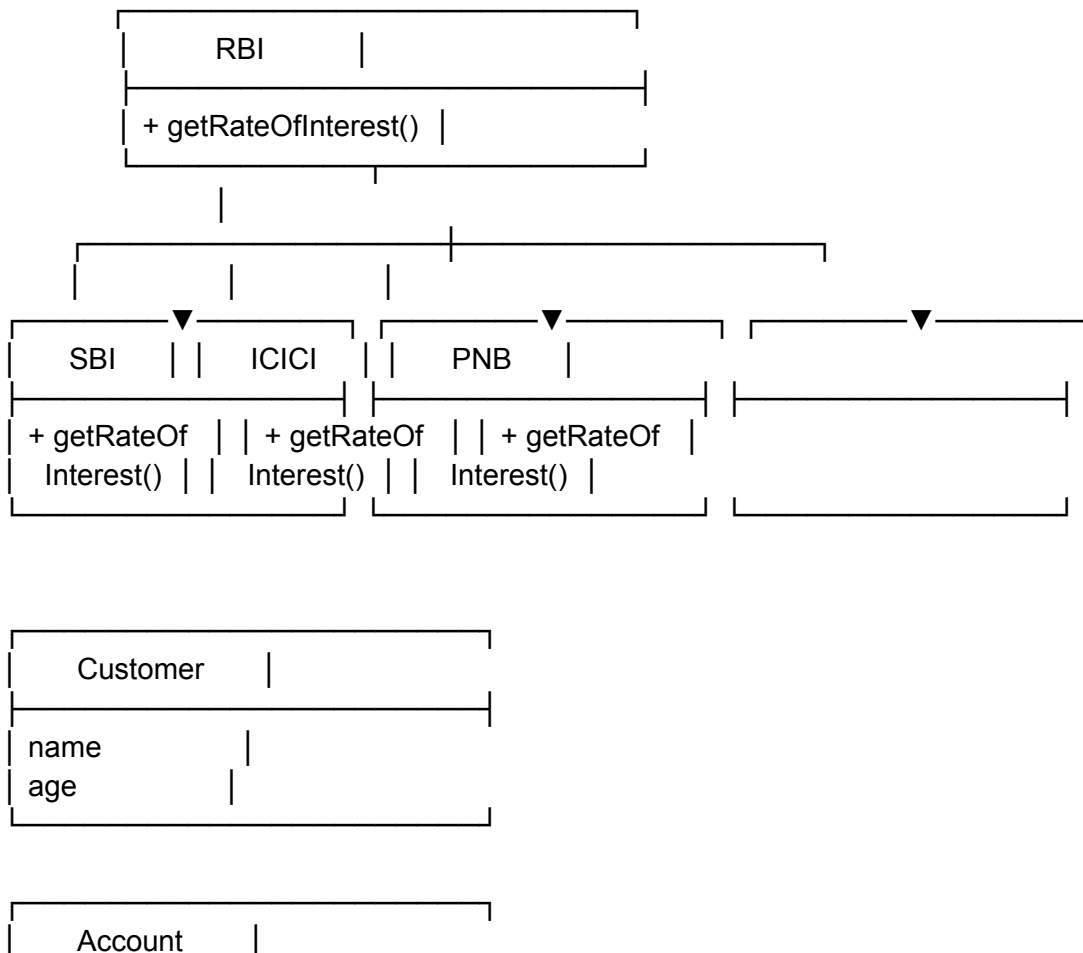
Algorithm / Logic Description

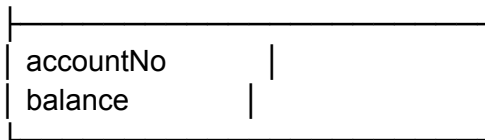
1. Create a **Customer** class with name and age attributes.
2. Create an **Account** class with account number and balance attributes.

3. Create a parent class named **RBI**.
4. Define the **getRateOfInterest()** method in the **RBI** class.
5. Create child classes named **SBI**, **ICICI**, and **PNB** that inherit from **RBI**.
6. Override the **getRateOfInterest()** method in each child class with its respective interest rate.
7. Create a reference variable of the parent class **RBI**.
8. Read the bank name from the user.
9. According to the entered bank name, create the corresponding child class object.
10. Store the child class object in the **RBI** reference variable.
11. Call **getRateOfInterest()** using the **RBI** reference.
12. Due to method overriding, the corresponding child class method is executed at runtime.
13. Display the rate of interest.



UML Class Diagram





Relationship: **SBI**, **ICICI**, and **PNB** inherit from the **RBI** class and override the `getRateOfInterest()` method.



Source Code

```
import java.util.Scanner;

// creating a class for customer
class Customer {

    String name;
    int age;

    // constructor
    Customer(String name, int age) {
        this.name = name;
        this.age = age;
    }
}

// creating a class for account
class Account {

    int accountNo;
    double balance;

    // constructor
    Account(int accountNo, double balance) {
        this.accountNo = accountNo;
        this.balance = balance;
    }
}

// creating parent class RBI
```

```
class RBI {

    // method to return rate of interest
    double getRateOfInterest() {
        return 4.0;
    }
}

// creating SBI class
class SBI extends RBI {

    // overriding getRateOfInterest method
    double getRateOfInterest() {
        return 7.0;
    }
}

// creating ICICI class
class ICICI extends RBI {

    // overriding getRateOfInterest method
    double getRateOfInterest() {
        return 6.5;
    }
}

// creating PNB class
class PNB extends RBI {

    // overriding getRateOfInterest method
    double getRateOfInterest() {
        return 6.0;
    }
}

// main class
public class Bank {

    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);

// taking bank name from user
System.out.print(
    "Enter the Bank name to find the rate of Interest : "
);

String name = sc.nextLine();

// creating parent class reference
RBI bank;

// creating SBI object
if (name.equalsIgnoreCase("SBI")) {
    bank = new SBI();
}

// creating ICICI object
else if (name.equalsIgnoreCase("ICICI")) {
    bank = new ICICI();
}

// creating PNB object
else if (name.equalsIgnoreCase("PNB")) {
    bank = new PNB();
}

// creating RBI object for other bank names
else {
    bank = new RBI();
}

// calling overridden method
System.out.println(
    name.toUpperCase() +
    " rate of interest is : " +
    bank.getRateOfInterest() + "%"
);

sc.close();
}
}
```



Compilation & Execution Commands

Save the source code exactly as:

Bank.java

Compile the program:

```
javac Bank.java
```

Run the program:

```
java Bank
```



Output Screen

Example 1: SBI

Enter the Bank name to find the rate of Interest : SBI
SBI rate of interest is : 7.0%

Example 2: ICICI

Enter the Bank name to find the rate of Interest : ICICI
ICICI rate of interest is : 6.5%

Example 3: PNB

Enter the Bank name to find the rate of Interest : PNB
PNB rate of interest is : 6.0%

Example 4: Other Bank

Enter the Bank name to find the rate of Interest : ABC
ABC rate of interest is : 4.0%

```
● PS C:\Users\navsee\OneDrive\Desktop\programs> javac Week7.java
● PS C:\Users\navsee\OneDrive\Desktop\programs> java Week7
Enter the Bank name to find the rate of Interest : sbi
SBI rate of interest is : 7.0%
● PS C:\Users\navsee\OneDrive\Desktop\programs> java Week7
Enter the Bank name to find the rate of Interest : cgi
CGI rate of interest is : 4.0%
● PS C:\Users\navsee\OneDrive\Desktop\programs> java Week7
Enter the Bank name to find the rate of Interest : ghj
GHJ rate of interest is : 4.0%
```